DP Graded Quiz M1

Question 1

(**Note:** Use Python to solve this question.) Compute the price of a European call with $\sigma = 0.5$, $T = 1$, $N = 1000$, $r = 0.1$, $K = 90$, and $S_0 = 100$.

- 32.41
- 29.98
- 28.65
- 28.54

Question 2

Consider a European Call Option with strike price (K), maturity T=2 years. Consider also a 2-step (N= 2) binomial tree with $S_0 = 50$, $u = 1.1$, and $d = 0.9$. Compute the value of K if the payoff of the call option for path {u,d} is 19.5?

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- 50
- 30
- 40
- 35

Question 3

(**Note:** Use Python to solve this question.)_ _Consider a 10-step binomial tree where, $u = 1.2$, $d = 0.8$, and the risk-free rate is equal to zero. Compute the delta, Δ_{ddddd}, at $t = 5$ for a call option with strike 90, $S_0 = 100$, maturity in 10 years and $N = 10$.

- 0.2852
- 0.0616
- 0.3656
- 0

Question 4

(**Note:** Use Python to solve this question.) Compute the price of a European call with $u = 1.2$, $d = 0.8$,

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$T = 1 = N$, $r = 0$, $K = 90$, and $S_0 = 100$.

- 25
- 13
- 5
- 15

Question 5

Q-measure makes reference to which of the following:

- risk-neutral and real-world probabilities
- risk-neutral probabilities
- volatility matching adjustment
- real-world probabilities

Question 6

Consider a European Call Option with strike price K= 80, maturity T= 4 years. Consider also a 4-step (N= 4) binomial tree with $S_0 = 100$, $u = 1.1$ and $d = 0.8$. What is

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the payoff of the call option for path {u,d,d,u}?

- 77.44
- 25
- 0
- 3.66

Question 7

Consider a 2-step binomial tree where the probability of upward movement is 0.5, $u = 1.2$, $d = 0.8$, and the risk-free rate is equal to zero. Compute the delta, Δ_u, at $t = 1$ for a put option with strike 74, $S_0 = 100$ and maturity in two years.

- $\Delta_u = 0.234$
- $\Delta_u = 1$
- $\Delta_u = 0.3125$
- $\Delta_u = 0$

Question 8

Consider a European Put Option with strike

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price $K = 90$, maturity $T = 1$ years. Consider also a 1-step ($N = 1$) binomial tree with $S_0 = 100$, $u = 1.1, d = 0.8$, and $r_f = 0$. Compute the amount (in shares) that we should invest (or short-sell) in the underlying asset today and the amount (in monetary units) the we need to borrow (or lend) to hedge the short position in the put.

- Buy 0.66 shares of the underlying asset and lend 36.66 monetary units.
- Buy 0.33 shares of the underlying asset and borrow 36.66 monetary units.
- Buy 0.33 shares of the underlying asset and lend 36.66 monetary units.

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QUESTIONS

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

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- Sell 0.33 shares of the underlying asset and lend 36.66 monetary units.

Question 9

Consider a binomial tree with $u = 1.2$ and $dt = 0.1$. Compute σ.

- $\sigma = 0.3$
- $\sigma = 0.58$
- $\sigma = 0.4512$
- $\sigma = 0.23$

Question 10

In a 2-step binomial tree stock prices at any point in time can make upward ($u = 1.1$) or downward (d) movements. The initial underlying asset price is 100. The price of the underlying asset is equal to 88 for the path {d, u}. What is the value of d?

- 0.9

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- 0.8
- 0.7
- 0.85

Question 11

Consider a 1-step binomial tree where the probability of upward movement is 0.5, $u = 1.3, d = 0.9$, and the risk-free rate is equal to zero. Compute the delta at $t = 0$ for a put option with strike 100, $S_0 = 100$ and maturity in one year.

- 0.5
- 0.25
- -0.5
- -0.25

Question 12

Consider a European Put Option with strike price $K = 100$, maturity $T = 1$ years. Consider also a 1-step ($N = 1$) binomial tree with $S_0 = 100$,

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$u = 1.2, d = 0.8$ and $r_f = 0.1$. Compute the amount (in shares) that we should invest (or short-sell) in the underlying asset today and the amount (in monetary units) the we need to borrow (or lend) to hedge the short position in the put.

- Sell 0.5 shares of the underlying asset and borrow 54.29 monetary units
- Sell 0.5 shares of the underlying asset and lend 54.29 monetary units
- Buy 0.5 shares of the underlying asset and lend 54.29 monetary units
- Sell 1 share of the underlying asset and borrow 54.29 monetary units

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Question 13

Consider a binomial tree with $\sigma = 0.3, r = 0.1$ and $dt = 0.01$. Compute p.

- $p = 0.5592$
- $p = 0.5292$
- $p = 0.4918$
- $p = 0.5092$

Question 14

Consider a 1-step binomial tree with $u = 1.1$ and $d = 0.8$ and a risk-free rate equal to 5%. Compute the risk-neutral probability of having a downward movement.

- 0.16
- 0.25
- 0.84
- 0.75

Question 15

Consider a 1-step binomial tree

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where the probability of upward movement is 0.5, $u = 1.3, d = 0.9$, and the risk-free rate is equal to zero. Compute the delta at $t = 0$ for a put option with strike 95, $S_0 = 100$ and maturity in one year.

- -0.125
- 0.25
- 1
- 0.125

Question 16

Consider a European Call Option with strike price $K = 60$, maturity $T = 1$ years. Consider also a 1-step ($N = 1$) binomial tree with $S_0 = 100$, $u = 1.1, d = 0.8$, and $r_f = 0$. Compute the amount (in shares) that we should invest (or short-sell) in the underlying asset today and the amount (in monetary units)

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the we need to borrow (or lend) to hedge the short position in the call.

- Short-sell 1 share of the underlying asset and lend 60 monetary units.
- Buy 1 share of the underlying asset and borrow 60 monetary units.
- Short-sell 1 share of the underlying asset and borrow 60 monetary units.
- Buy 1 share of the underlying asset and lend 60 monetary units.

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